

Laser Type:	ND-YAG
Wavelength (nm):	1064
Beam Diameter (mm):	5
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Mode:	continuous-wave
Pulse Duration (sec):	
Pulse Frequency (Hz):	
Average Joules/Pulse (J):	
Maximum Joules/Pulse:	
Average Power (watts):	2
Maximum Power (watts):	10
Laser Type #1:	ND-YAG
Operation Power (W or J):	2
Wavelength (nm):	1064
Wavelength Attenuated:	
Optical Density Required:	4
Eyewear (Mfg/model):	Thorlabs LG1
When use is required:	When adjusting
Additional Information:	At the beginning of each run, a thorough survey will be conducted to verify that all unwanted beams have been properly terminated.
General notes:	• Suitable eye protection is needed during alignment and microscope adjustment. • Ensure that laser beam does not pass through areas of heavy traffic and/or use. • Make sure reflective objects, e.g. metal rings and watches, are not worn in the vicinity of the laser. • Laser beam is fully enclosed in a closed system with an interlock in place during normal operational use.
Protection notes:	• Protect against fumes and vapors by ensuring that all vents and airways in the room are kept open and unobstructed. • Electronic circuits should be properly grounded. • Compressed gas canisters should be securely fastened to the wall.

- Dyes should be handled according to their reactivity and toxicity; the workspace should be kept suitably ventilated and gloves should always be worn when handling dyes.
- Flammable chemicals should always be stored in the flammables cabinet until ready for use.
- The floor should be kept clean to minimize the risk of tripping.
- Sharp objects and dangerous chemicals should be stored safely within the appropriate storage compartment.

Operating notes:

General Operating Procedure

- Put on laser safety goggles if doing any adjustment or beam collimation operations
- Turn on the lab laser safety light
- Close laser lab entrance curtains
- During normal operational use, the laser will be in a completely enclosed system. An interlock in place will turn off the ND-YAG laser if the optical path is exposed to the user.

Alignment and Calibration

- Alignment of the ND-YAG will be performed by Lumicks Customer Support.
- Users will visualize the back-focal plane to check condenser-sample approach and beam collimation. If adjustments to the condenser height is required or interlock is over-ridden, suitable eye protection is necessary.

Start up and Operation

- Ensure that the laser is in a completely enclosed system and that the interlock is in place that will turn off the ND-YAG laser if the optical path is exposed to the user.
- If those controls are in place, then protective eyewear is not required.
- Turn on key at laser control box
- Laser takes about ~5 minutes to stabilize after initial turn-on

Shutdown

- Turn off the laser
- Turn off the laser controls
- Note: the interlock in place will turn off the ND-YAG laser if the optical path is exposed to the user during the shutdown process.